Badri Narayanan Narasimhan, Ph.D.

Postdoctoral Scholar | University of California, San Diego

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PROFESSIONAL SUMMARY

Postdoctoral scholar investigating how extracellular matrix (ECM) remodeling regulates cellular organization, mechanosensation, and cancer progression. My research integrates engineered biomaterials, mechanobiology, quantitative modeling, and spatial profiling to define how matrix mechanics, adhesion, and degradation shape cellular states. Building on my training in materials engineering and my work establishing links between matrix degradation and stress-relaxation (PNAS 2025), I am developing predictive experimental frameworks that connect engineered microenvironments with cancer organoids and patient tissues. My emerging research program investigates how dynamic ECM remodeling regulates immune-interface remodeling and thereby shapes cancer-immune cell interactions. My long-term goal is to establish an independent research program that decodes and reconstructs dynamic cell-ECM interactions in cancer and other epithelial diseases.

EDUCATION

- **Postdoctoral Scholar in Bioengineering**, UC San Diego, USA (07/2022 – Present)
- **Ph.D. in Chemical & Materials Engineering**, University of Auckland, New Zealand (03/2018 – 06/2022)
- **M.S. in Nano & Radio Sciences** (Minor: Micro & Nano Sciences), Aalto University, Finland (09/2014 – 04/2017)
- **B.E. in Electronics & Communication Engineering** | Mepco Schlenk Engineering College, India (05/2006 – 04/2010)

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Postdoctoral Scholar, University of California, San Diego (Supervisor: Prof. Stephanie Fraley) (2022 – Present)

- Investigating cellular mechanical sensing of matrix degradation for cancer organoids design.
- Collaborated with theorists to demonstrate that ECM confinement regulates rotational dynamics of single cells (PNAS, 2026, Accepted)
- Demonstrated that matrix degradation enhances stress relaxation, regulating cell adhesion and spreading (*PNAS*, 2025).
- Demonstrated that collagen fibril architecture dictates matrix degradability (*Soft Matter*, 2024).

Graduate Researcher (Ph.D.), University of Auckland (Supervisor: Prof. Jenny Malmström) (2018 – 2022)

- Developed tunable synthetic hydrogels for mechanotransduction with independently tunable elastic and viscous properties.
- Evaluated structure-property relationships using rheology, neutron scattering, and in-depth mechanical characterization tools.
- Optimized synthetic hydrogel environments for advanced cell culture applications.

Research Assistant & Master Thesis, Aalto University, Finland (Supervisor: Prof. Olli Ikkala) (2015 – 2016) Developed cellulose nanocrystal-functionalized surfaces and nanocellulose hydrogels with controlled mechanical and gelation properties.

Design Engineer, Spectrum Solutions & Technologies Pvt. Ltd (2011-2013)

Programmer analyst Trainee, Cognizant Technologies Pvt. Ltd (2010-2011)

MENTORSHIP & TEACHING

- **Guest Lecturer**, UC San Diego: Taught graduate-level Mass Transfer, covering lectures on Reaction diffusion systems and Cellular Potts models (2025).
- **Research Mentor**, UC San Diego: Currently guiding 3 undergraduate students and guided and mentored 4 graduate students.
- **Thesis Advisor**, University of Auckland: Mentored 2 undergraduate theses on hydrogel adhesion properties (2021).
- **Lab Course Instructor (CM204)**, University of Auckland: Chemical and Materials Engineering lab (2019).
- **Math Tutor**, India: Provided college-level math tutoring for a visually challenged student (2013).

HONORS, SERVICE & PROFESSIONAL DEVELOPMENT

- **Patent:** Provisional Patent (2025): "Methods and systems for controlling cell migration"
- **Early Career Editorial Board member:** *Biomimetics,* 2026.
- **Guest Editor Assistant:** "*Advances in Tissue Engineering and Next-Generation Biomaterials: From Techniques to Emerging Applications*", *Bioengineering,* 2026.
- **Judge:** Lab Expo Poster Presentation, UC San Diego (2025, 2026).
- **Speaker:** ACS Symposium (2025), Biophysical Society (2025), Rutgers Physical Chemistry Seminar (2021).
- **Peer Reviewer:** 29 reviews across 7 journals, including *Advanced Functional Materials, Acta Biomaterialia, Small,* and *International Journal of Biological Macromolecules*. (ORCID: 0000-0002-2936-0398).
- **COMPASS Postdoctoral Training Program** (WashU Medicine) - Certificate of Completion, 2026 *(NIH-supported program focused on leadership, mentoring, and career development for postdoctoral researchers)*

PUBLICATIONS

PEER-REVIEWED PUBLICATIONS & ACCEPTED MANUSCRIPTS

1. Echeverria-Alar, S., **Narasimhan, B. N.**, et al. (2026). Single-cell chiral symmetry breaking under confinement. *Proceedings of the National Academy of Sciences*. Accepted. Preprint: *bioRxiv*.
2. **Narasimhan, B. N.**, & Fraley, S. I. (2025). Matrix degradation enhances stress relaxation, regulating cell adhesion and spreading. *Proceedings of the National Academy of Sciences*, 122(13), e2416771122.
3. Debnath, B., **Narasimhan, B. N.**, et al. (2024). Modeling collagen fibril degradation as a function of matrix microarchitecture. *Soft Matter*, 20, 9286-9300.
4. **Narasimhan, B. N.**, et al. (2023). Hydrogen bonding dissipating hydrogels: Network structure design and property relationships. *Journal of Colloid and Interface Science*, 630, 638-653.

5. Ryssy, J., Lehtonen, A. J., Loo, J., Nguyen, M.-K., Seitsonen, J., Huang, Y., **Narasimhan, B. N.**, Pokki, J., Kuzyk, A., & Manuguri, S. (2022). DNA-engineered hydrogels with light-adaptive plasmonic responses. *Advanced Functional Materials*, 32(37), 2201249.
6. Ting, M. S., Vella, J., Raos, B. J., **Narasimhan, B. N.**, Svirskis, D., Travas-Sejdic, J., & Malmström, J. (2022). Conducting polymer hydrogels with electrically-tuneable mechanical properties as dynamic cell culture substrates. *Biomaterials Advances*, 134, 112559.
7. **Narasimhan, B. N.**, et al. (2021). Hydrogels with tunable physical cues in cellular mechanotransduction. *Advanced NanoBiomed Research*, 1(10), 2100059.
8. Ting, M. S., **Narasimhan, B. N.**, Travas-Sejdic, J., & Malmström, J. (2021). Soft conducting polymer polypyrrole actuation based on poly(N-isopropylacrylamide) hydrogels. *Sensors and Actuators B: Chemical*, 343, 130167.
9. Manuguri, S., van der Heijden, N. J., Nam, S. J., **Narasimhan, B. N.**, Wei, B., Cabero Z, M. A., ... & Malmström, J. (2021). Polymer micelle directed magnetic cargo assemblies towards spin-wave manipulation. *Advanced Materials Interfaces*, 8(15), 2100455.
10. **Narasimhan, B. N.**, et al. (2021). A comparative study of tough hydrogen bonding dissipating hydrogels made with different network structures. *Nanoscale Advances*, 3(10), 2934-2947.
11. **Narasimhan, B. N.**, et al. (2020). Mechanical characterization for cellular mechanobiology: Current trends and future prospects. *Frontiers in Bioengineering and Biotechnology*, 8, 595978.

MANUSCRIPTS IN REVISION

12. Echeverria-Alar, S., **Narasimhan, B. N.**, et al. (2026). Lumens as active balloons: A biological physical review. *Reports on Progress in Physics*. In revision. Preprint: *bioRxiv*.
13. Jimenez, J. M., **Narasimhan, B. N.**, & Fraley, S. I. (2026). Transient calcium signaling uncouples YAP activity from matrix constraints to drive context-dependent transcriptional outputs. *Acta Biomaterialia*. In revision.
14. DeWitt, J. T., Jimenez-Tovar, D., Nguyen, C., Oropeza, E., Raghunathan, M., Karabay, E. T., Lamichhane, A., Gabel, B., **Narasimhan, B. N.**, Candy, F., Kuwahara, L., Head, A. G., Batalov, S., Kirk, E., Zare, S., Bainbridge, M., Fraley, S. I., Raghavan, S. A., Katira, P., Luna Lopez, C., Troester, M. A., Freedland, S. J., Lui, A., & Haricharan, S. (2026). The impact of patient biology on the breast tumor microenvironment. *Cancer Research*. In revision.

MANUSCRIPTS IN PREPARATION

15. **Narasimhan, B. N.**, Echeverria-Alar, S., Ding, S., Ivaturi, R., Zhang, D., & Fraley, S. I. (2026). Interactions between matrix degradability and adhesion presentation are sufficient to recapitulate distinct cancer phenotypes. *Manuscript in preparation.*
16. **Narasimhan, B. N.**, & Fraley, S. I. (2026). Spatial transcriptomic and proteomic profiling reveals molecular transitions from MGA to AMGA and invasive carcinoma. *Manuscript in preparation.*
17. Rowell, M., Karabay, E., Ding, S., **Narasimhan, B. N.**, Katira, P., & Fraley, S. I. (2026). Combining persistent random and oscillatory walk models captures the influence of confinement on 3D epithelial migration. *Manuscript in preparation.*

BOOK CHAPTER

18. Manuguri, S., & **Narasimhan, B. N.**, et al. (2023). Forms of graphene IV-functionalized graphene. In *Recent Advances in Graphene and Graphene-Based Technologies.* Bristol, UK: IOP Publishing

REFERENCES

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